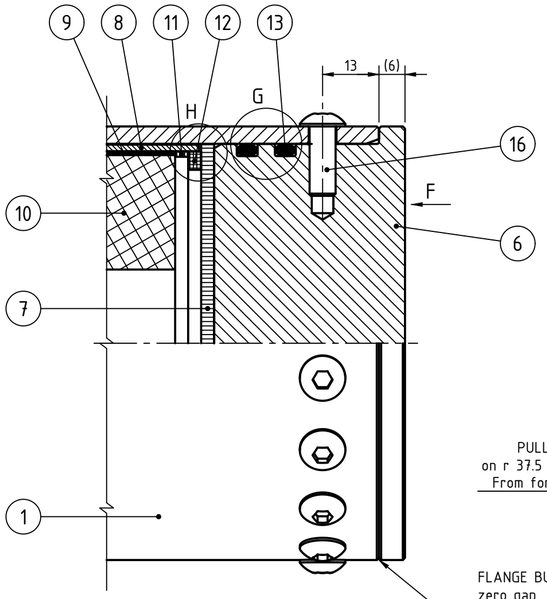
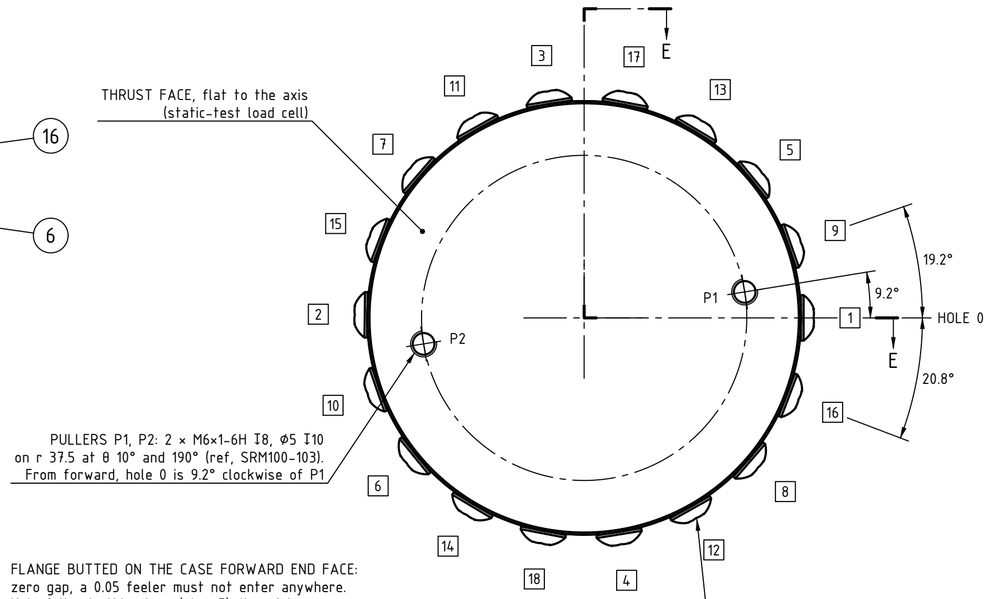


E–E (1:1)



PULLERS P1, P2: 2 × M6×1-6H ↧8, ⌀5 ↧10 on r 37.5 at θ 10° and 190° (ref, SRM100-103). From forward, hole 0 is 9.2° clockwise of P1

FLANGE BUTTED ON THE CASE FORWARD END FACE: zero gap, a 0.05 feeler must not enter anywhere. Hole 0 lies in this plane (view F). Item 6 is a serialised pair with item 1; clocking: note 12.

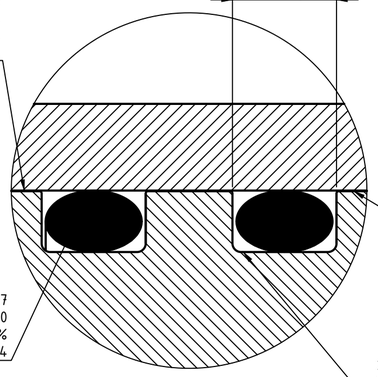
F (1:1)

18 × item 16, M6×16-010.9 (ISO 7380-1), 4.0–6.0 N·m, Loctite 243. Order: the boxed numbers, 3 passes (note 13). Holes in item 1: ⌀6.40 +0.03/0; 12 engaged in item 6 (ref)

LOADING AND FORWARD END CLOSING

1. Item numbers and materials: parts list, sheet 1. Tooling (casting mandrel, hole plugs, clamp collar, puller bridge) is not in this drawing set.
2. Aft end first (sheet 2); first build and proof: sheet 2, note 2. Bond item 7 to item 6 only after the proof (run with no insulators or liner).
3. Bond item 7 to item 6's inboard face (README assembly step 1, SRM100-107): abrade, IPA wipe, silicone primer SS4004P, RTV106 or equal over the whole face, bondline 0.1–0.2, OD centred on the land within 0.1; cure 24 h. Then measure t, flange bearing face to item 7's aft face, at 4 points, mean (nominal 41.00 + bondline).
4. Lube (O-rings, liner OD, insulator edges): Parker Super O-Lube or DC-55, thin film. Parts replaced per note 17 (labels included) are always new.
5. Grains are cast in their casting tubes (item 9) on the casting mandrel; the tube stays on the grain. Mandrel out, measure and record every grain length (147.20 ±0.10) and spacer (3.00 ±0.02): stack-up table.
6. Cavity and liner: with the aft end closed, measure the depth d_N from the case forward end face to item 4's forward face (nominal 647.80). Cavity C = d_N − t (nominal 606.80). Cut item 8 to C − 0.50 0/−0.10 (SRM100-110), lubricate its OD (note 4) and slide it in by hand until it bears on item 4.
7. Load by hand, aft to forward: S0 on item 4, G1, S1, G2, S2, G3, S3, G4, S4 (stack view E–E). Each tube and spacer must slide freely in the liner.
8. Liner float, in place of the README feeler check (the gap is closed off once item 6 is in): depth d_L from the case forward end face to the liner's forward end ≥ t + 0.30 at 4 points (nominal 41.50 for t = 41.00).
9. Take-up gap: depth d_S from the case forward end face to the top of S4 at 4 points, mean; g = d_S − t (nominal 44.00 − 41.00 = 3.00; ±0.50 stack, ±0.10 case length). Measure the free thickness t_s of item 12 (4 points, mean) and accept 0.50 ≤ g / t_s ≤ 0.75 (25–50 % compression; nominal 37.5 %; SRM100-108): sheet 3.2 for g 1.60–2.40, 4.8 for g 2.40–3.60, 6.4 for g 3.20–4.80. Lay item 12 on S4.
10. Hole plugs: push 18 acetal plugs into the case's forward-end holes before item 6 goes in (as sheet 2, note 6).
11. Fit 2 × item 13, FKM 90A (Parker V0709-90 or equal), in the grooves of item 6 (detail G). Item 6 is cut to the forward-zone bore A recorded on SRM100-101 (92.000 ≤ A ≤ 92.070): land A − 0.06 0/−0.02, groove bottoms A − 5.67 0/−0.04. Detail G is drawn at A = 92.000.
12. Clocking (view F): enter item 6 with hole 0 on the case's hole 0, the scribed clocking line straight across case end and flange, the FWD markings of items 1 and 6 (SRM100-101 note 8, SRM100-103 note 11) side by side. Check: seen from forward, hole 0 is 9.2° clockwise of puller P1 (θ 10°). Serialised pair: never fit another closure. Push in by hand until the flange butts on the case end face (zero gap).
13. Pull each plug as its screw goes in. Screws 18 × item 16 with Loctite 243, in the boxed order of view F (opposite pairs, criss-cross): pass 1 to about 2.0 N·m, pass 2 to 4.0–6.0 N·m, pass 3 at final torque hole by hole round the circle, as a check. Never an M6×10 (item 15) here: 1 d engagement only.
14. No pressure port (sheet 1 note 11): item 6 is solid (no port, no pocket) and item 7 is a full disc; nothing is fitted or plugged at the forward end.
15. Igniter last, at the stand; cap per sheet 2 note 11. It goes up the throat about 615 mm from the nozzle exit, to the middle of G4's core (z 570.2).
16. Extraction, labels read first (sheet 2 note 12), never lever or knock: screws out; split clamp collar (100.0 bore) on the case OD at least 50 mm from the forward end face, clear of the screw heads at 13 (sweep ⌀106.04 ref) and the marking at 20–45 from that face; puller bridge, 2 × M6 draw screws in item 6's P1, P2 (view F), or react on the stand's thrust plate.
17. After every firing: replace items 4, 5, 7, 8, 9, 10, 11, 12, 13, 14, all 36 screws (items 15, 16) and the temperature labels. Re-use items 1, 2 and 6 after inspection (SRM100-101 R10 for the case), and item 3 while its throat is ≤ ⌀23.9 and it is uncracked. Scrape item 7 and the old RTV off item 6 with a plastic scraper (SRM100-103); re-gauge every bolt hole against its baseline (SRM100-101 R10).

BORE, item 1: ⌀A +0.030/0, honed Ra 0.4 (0.8 max), circumferential lay; 92.000 ≤ A ≤ 92.070, recorded per zone (SRM100-101). Drawn at A = 92.000.



LAND, item 6: ⌀A − 0.06 0/−0.02 (⌀91.94 at A = 92.000) clearance on ⌀ 0.06–0.11 (0.030–0.055 radial)

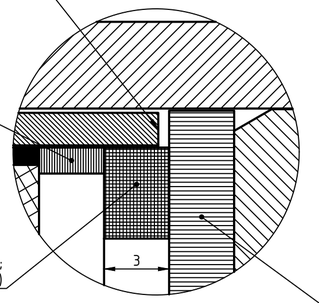
2 × item 13, AS568-237 FKM 90A (Parker V0709-90 or equal), squeeze 19.7 % lube: note 4

2 GROOVES ⌀A − 5.67 0/−0.04 (⌀86.33 at A = 92.000) gland depth 2.835 (ref)

G (5:1)

LINER END, item 8: floats 0.50 short of item 7 (0.30 min, note 8)

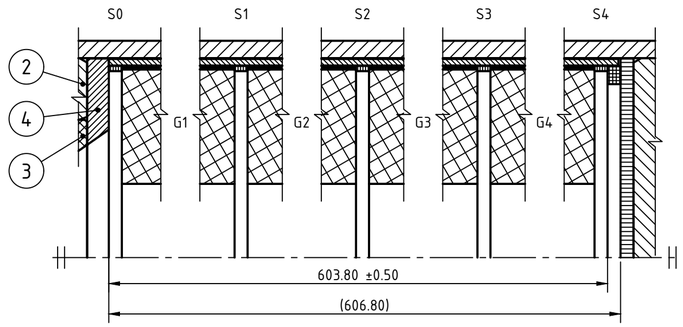
S4, item 11, 3.00 ±0.02



item 12, 4.80 free, 3.00 installed (37.5 %); 27.1–47.9 % over the ±0.50 stack (note 9)

item 7, EPDM 3.0 ±0.2, RTV-bonded to item 6 (note 3); the 0.1–0.2 bondline is not in the model

H (5:1)



E–E (1:1), GRAIN STACK – INTERRUPTED VIEW

GRAIN STACK AXIAL STACK-UP (nominal; z from the case aft end face)

POS	ELEMENT	ITEM	LENGTH	TOL	z AFT	z FWD
S0	Spacer 0 (on item 4)	11	3.00	±0.02	43.00	46.00
G1	Grain 1 in its casting tube	10, 9	147.20	±0.10	46.00	193.20
S1	Spacer 1	11	3.00	±0.02	193.20	196.20
G2	Grain 2 in its casting tube	10, 9	147.20	±0.10	196.20	343.40
S2	Spacer 2	11	3.00	±0.02	343.40	346.40
G3	Grain 3 in its casting tube	10, 9	147.20	±0.10	346.40	493.60
S3	Spacer 3	11	3.00	±0.02	493.60	496.60
G4	Grain 4 in its casting tube	10, 9	147.20	±0.10	496.60	643.80
S4	Spacer 4	11	3.00	±0.02	643.80	646.80
STACK	5 spacers + 4 grains (worst case)	–	603.80	±0.50	43.00	646.80
T	Take-up ring, installed (4.80 free)	12	3.00	= g	646.80	649.80
C	Cavity = stack + take-up	–	(606.80)	meas.	43.00	649.80
L	Liner, cut to C − 0.50	8	606.30	0/−0.10	43.00	649.30
–	Liner float to item 7	–	0.50	0.30 min	649.30	649.80
I	Forward insulator, EPDM, bonded	7	3.00	±0.20	649.80	652.80
P	Forward closure, plug	6	38.00	gen.	652.80	690.80
F	Forward closure, flange	6	6.00	gen.	690.80	696.80
–	Case (aft end face z 0)	1	690.80	±0.10	0.00	690.80

Stations (ref): O-ring grooves 660.40 / 669.20, bolt row 677.80, thrust face 696.80.
Motor overall 749.51 (cap back face z −52.71 to the thrust face). Igniter tip station z 570.2 (note 15).
TOL gen. = ISO 2768-mK (SRM100-103), case per SRM100-101; the build uses the measured t, d_N and d_S.

GAUGING RECORD AT LOADING (one per firing, file with the motor log)

QUANTITY (notes 3–9, 12, 13)	NOMINAL	ACCEPT	MEASURED
t: item 6 flange face to item 7, 4 pts	41.00	+ bondline	
d_N: case end face to item 4, 4 pts	647.80	record	
C = d_N − t: cavity	606.80	record	
L: liner cut length	606.30	C − 0.50 0/−0.10	
Stack: measured spacers + grains	603.80	±0.50	
d_L: case end face to liner end, 4 pts	41.50	≥ t + 0.30	
d_S: case end face to S4 top, 4 pts	44.00	record	
g = d_S − t: take-up gap	3.00	record	
t_s: item 12 free thickness, 4 pts	4.80	sheet per g	
g / t_s	0.625	0.50–0.75	
Flange gap to the case end face	0	0.05 feeler: no entry	
Screw torque, final pass	4.0–6.0 N·m	3 passes	

USED ON: SRM-100 motor (Helios) – static test and flight

MATERIAL: See parts list, sheet 1	MASS: 10.60 kg	GENERAL TOLERANCES: ISO 2768-mK	SURFACE TEXTURE: –	EDGES: –
DOCUMENT TYPE: ASSEMBLY DRAWING	DOCUMENT STATUS: NOT CHECKED – FOR REVIEW	SCALE: 1:1	PROJECTION	UNITS: mm
LEGAL OWNER: HELIOS SRM-100 SOLID ROCKET MOTOR	TITLE, SUPPLEMENTARY TITLE: MOTOR ASSEMBLY – FORWARD END — Forward closure build-up, grain stack loading and stack-up, tightening			
CREATED BY: text-to-cad / Claude	APPROVED BY: –	DATE OF ISSUE: 2026-09-23	IDENTIFICATION NUMBER: SRM100-000	REV: A — SHEET: 3 / 3